

The vertex centroid of focal antipedals of even elliptic billiards

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Abstract

For a periodic billiard in an ellipse with a strictly nested confocal elliptical caustic and even least period, we prove that the unweighted vertex centroids of the original antipedals about the center and the two foci are constant throughout its Poncelet family. The central centroid is zero; an explicit focal coefficient depends only on the axes, caustic parameter, period and perimeter. This answers invariant k405 as posed without proof by Reznik, Garcia and Koiller. The proof uses opposite edges, an elementary finite circle action, and perimeter stationarity. Ferudun's subsequent deposited note states the same theorem and coefficient and uses the same pair-and-moment mechanism; we explicitly credit that overlap and make no absolute priority or independent-discovery claim.

1 Statement, source and conventions

Let $a > b > 0$, $c = \sqrt{a^2 - b^2}$, and consider

$$E : \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1, \quad E_\lambda : \frac{x^2}{a^2 - \lambda} + \frac{y^2}{b^2 - \lambda} = 1, \quad 0 < \lambda < b^2.$$

Write $O = (0, 0)$ and $F_\sigma = (\sigma c, 0)$, $\sigma = \pm 1$. A periodic billiard tangent to E_λ extends to the smooth Poncelet family with the same caustic; we use this classical closure theorem in its usual confocal-ellipse form [5]. For successive billiard vertices A, B and a pole M , define $Q_M(A, B)$ as the intersection of the full lines through A and B perpendicular to $A - M$ and $B - M$, respectively. For an orbit of least period N , its *antipedal vertex centroid* is

$$\mathcal{C}_M = \frac{1}{N} \sum_{i=0}^{N-1} Q_M(P_i, P_{i+1}), \quad P_N = P_0. \tag{1}$$

This is a vertex mean, not an area centroid or a mean of pedal feet. The antipedal belongs to the original orbit, not to its outer tangent polygon. All three poles lie strictly inside E_λ , so its tangent chords do not pass through them. The intersections in (1) are therefore finite and unique.

Theorem 1. *Suppose the oriented billiard has even least period N (hence $N \geq 4$) and caustic E_λ . Let L be the perimeter of its Poncelet family, and put*

$$K = a^2b^2 - \lambda c^2, \quad H = \frac{a^2}{c^2} \left(1 - \frac{bL}{2a\sqrt{\lambda}N} \right).$$

Then throughout that family,

$$\mathcal{C}_O = (0, 0), \quad \mathcal{C}_{F_\sigma} = \left(\sigma c \left[-1 + \frac{KH}{a^2(b^2 - \lambda)} \right], 0 \right). \quad (2)$$

Both simple and primitive star orbits, and either traversal orientation, are allowed.

The source is k405 in Table 5 of [1], p. 348, and of [2], printed p. 7. It specifies even N and poles O, F_1, F_2 in the confocal-ellipse setting, with proof and value left as questions. The source does not formally define least period. Its use of opposite vertices under central inversion in [2, Section 3.7] supports the primitive-period interpretation used explicitly here. Repeating an even-period polygon leaves (1) unchanged; repeating an odd polygon an even number of times does not put it under this theorem. Degenerate two-bounce orbits, limiting degenerate caustics and hyperbolic caustics are outside the statement. In a circle the foci coincide with O and the conclusion for even least period is zero by central symmetry; the coefficient above, which divides by c^2 , is not used for a circle.

The least-period restriction is substantive. For $a = 2, b = 1$, the billiard triangle with vertices $(2, 0)$ and

$$\left(\frac{2(1 - \sqrt{13})}{3}, \pm \frac{\sqrt{2\sqrt{13} - 5}}{3} \right)$$

has caustic $\lambda = (8\sqrt{13} - 20)/9$ and origin antipedal centroid $((35 - 5\sqrt{13})/24, 0) \neq 0$; the reflected triangle has the opposite centroid. Repeating each twice gives six-entry lists with these same centroids. Exact reflection, tangency and centroid checks for this boundary example are supplied in the supplement.

2 Opposite vertices and two chord moments

Lemma 1. *For the hypotheses of Theorem 1, $P_{i+N/2} = -P_i$.*

Proof. Let T be the Poncelet tangent map, choosing the directed chord with the inner ellipse on its left. This is a smooth orientation-preserving homeomorphism of the outer oval; the other tangent gives its inverse. The Poncelet family premise gives $T^N = \text{id}$, of order exactly N . Central inversion $S(P) = -P$ preserves the chosen side of a directed chord, so S commutes with T .

The finite group generated by S and T preserves the average of the pushforwards of arc-length probability measure over this group. This measure is nonatomic and positive on every nonempty arc. Its cumulative coordinate conjugates every group element to a rotation: an oriented measure-preserving map preserves all arc lengths in this coordinate. A rotation group has a unique element of order two. Both S and $T^{N/2}$ have order two, so they agree. $\square$

Opposite antipedal intersections for the pole O are Q and $-Q$. Lemma 1 already proves $\mathcal{C}_O = 0$.

Choose the orientation with the caustic to the left. Since O is inside the caustic, $\det(A, B) > 0$ for each directed edge. Lift its eccentric endpoint angles as $s - \delta, s + \delta$, with $0 < \delta < \pi/2$, and write

$$C = \cos s, \quad S = \sin s, \quad U = \cos \delta, \quad V = \sin \delta, \quad D^2 = \frac{C^2}{a^2} + \frac{S^2}{b^2}.$$

Then

$$\begin{aligned} A &= (a(CU + SV), b(SU - CV)), \\ B &= (a(CU - SV), b(SU + CV)). \end{aligned}$$

The chord line is $xC/a + yS/b = U$. Tangency to E_λ gives

$$U^2 = 1 - \lambda D^2, \quad V^2 = \lambda D^2. \quad (3)$$

Its positive length is

$$\ell = 2V\sqrt{a^2S^2 + b^2C^2} = 2ab\sqrt{\lambda}D^2. \quad (4)$$

Lemma 2. *The perimeter L is constant on the Poncelet family, and its chord coordinates satisfy*

$$\frac{1}{N} \sum_i C_i^2 = H, \quad \sum_i S_i C_i = 0. \quad (5)$$

Proof. Under independent tangent displacements of the billiard vertices, the first variation of perimeter is

$$dL = \sum_i (u_{i-1} - u_i) \cdot dP_i, \quad u_i = \frac{P_{i+1} - P_i}{|P_{i+1} - P_i|}.$$

The reflection law makes $u_{i-1} - u_i$ normal to E , so $dL = 0$. This calculation does not require the polygon to be simple. Along the smooth Poncelet family it proves that L has zero derivative and is constant on the connected circle of initial vertices.

Summing (4) and using $D^2 = b^{-2} - c^2 C^2 / (a^2 b^2)$ yields the first identity in (5). For the second, use the allowed comparison deformation $t_i \mapsto t_i + \epsilon$ of all eccentric vertex angles. It keeps each δ_i fixed and changes s_i to $s_i + \epsilon$. The deformed chords need not preserve the caustic. Differentiate the first expression in (4) *before* imposing a caustic constraint on the deformation. At the original billiard,

$$\left. \frac{d\ell_i}{d\epsilon} \right|_0 = \frac{2V_i c^2 S_i C_i}{\sqrt{a^2 S_i^2 + b^2 C_i^2}} = \frac{2\sqrt{\lambda} c^2}{ab} S_i C_i.$$

Stationarity of L gives the claimed sum, since $c, \lambda > 0$. □

3 The opposite-edge antipedal calculation

For a focus $F = (h, 0)$, $h = \pm c$, set

$$m = (aCU, bSU), \quad v = (-aSV, bCV), \quad d = m - F, \quad q = Q_F(A, B) - F.$$

Thus $A = m - v$ and $B = m + v$. Adding and subtracting the equations of the two antipedal lines gives

$$q \cdot d = R := |d|^2 + |v|^2, \quad q \cdot v = T := 2d \cdot v. \quad (6)$$

Its determinant is $bV(aU - hC) > 0$. Indeed,

$$U^2 - \frac{c^2 C^2}{a^2} = (b^2 - \lambda)D^2 > 0, \quad (7)$$

so $aU > |hC|$.

For clarity, Cramer's rule gives

$$q_x = \frac{bCVR - bSUT}{bV(aU - hC)}, \quad q_y = \frac{(aCU - h)T + aSVR}{bV(aU - hC)}.$$

For the opposite edge replace (m, v) by $(-m, -v)$ in (6), while keeping F fixed. Adding the two resulting absolute intersections, with $h^2 = c^2$, gives

$$\begin{aligned} (Q_F(A, B) + Q_F(-A, -B))_x &= \frac{2hb^2(C^2 - U^2)}{b^2 - \lambda}, \\ (Q_F(A, B) + Q_F(-A, -B))_y &= \frac{-2CSah[c^2(C^2 - U^2) - b^2]}{ba^2(U^2 - c^2C^2/a^2)}. \end{aligned} \quad (8)$$

Here the first expression has used (7) and $D^2 = C^2/a^2 + S^2/b^2$. The remaining simplifications are

$$c^2(C^2 - U^2) - b^2 = -KD^2, \quad b^2(C^2 - U^2) = -(b^2 - \lambda) + \frac{KC^2}{a^2}.$$

Consequently

$$Q_F(A, B) + Q_F(-A, -B) = \left(-2h + \frac{2hKC^2}{a^2(b^2 - \lambda)}, \frac{2hKSC}{ab(b^2 - \lambda)} \right). \quad (9)$$

This local identity uses tangency but not closure or the reflection law. By Lemma 1, summing its left side over all edges counts each antipedal intersection twice. Divide by $2N$ and apply Lemma 2. This proves (2). Reversing traversal only permutes the same full-line intersections, so it leaves the result unchanged.

4 Provenance, overlap and verification

Central symmetry, perimeter stationarity and confocal billiard conservation are classical; see, for example, [3, 5]. The arguments needed here have been supplied in full. Earlier related pedal-centroid and harmonic results in [3], together with the classical skew-hodograph correspondence discussed in [4], yield another reduction once the target-specific pair identity (9) is inserted. These ingredients alone are not a located earlier proof of the focal antipedal theorem. Our bounded audit of primary literature did not establish an earlier complete answer to k405; it does not establish absolute priority.

A proof was publicly posted in PR140 on September 30, 2026 [6]. Its archived original already contains the theorem and the pair formula. Ferudun's *Original Antipedal Centroids in Elliptic Billiards* [7] states the same scoped theorem and the same explicit coefficient. Its specific Zenodo record was created on October 1, 2026 at 23:58:07 UTC, with publication date October 2. Its proof also combines opposite-edge antipedal pairs with chord moments; a change to unit-normal coordinates gives the same identities. We credit this exact overlap. The documented dates concern those specific public records; they do not establish absolute priority, independent discovery, or any inference of copying or collaboration. The present note records a verified self-contained resolution of a question posed without proof in the 2020/2021 source, rather than claiming that the question remains unsolved at this note's publication.

The accompanying verification package contains explicit-guard rational and symbolic checks of the local line systems, tangency and pair identities, length differentiation, and finite-rotation parity controls. Its finite computations support reproducibility; they do not prove Poncelet closure for arbitrary parameters or replace the all-period arguments above. Separate adversarial audits tested genuine elliptic orbits, parameter boundaries, repeated odd polygons and deliberately false identities. Audit results and limits are described in the supplement.

AI assistance and review status. AI tools were used extensively in developing, drafting, reproducing and adversarially auditing this work. Independent AI review passes are not conventional human peer review. This preprint is unrefereed and has not undergone conventional human peer review at publication. The proof and supplementary programs are supplied for independent checking.

References

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